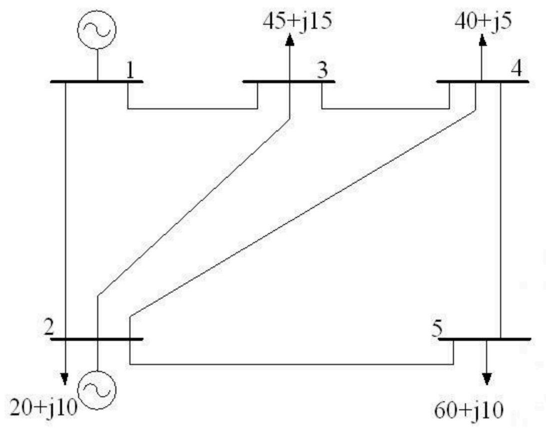
Python in Electrical Engineering

PandaPower

You are given a **5-bus test network**

1. **Create the power system model** in PandaPower using the specifications below.
2. **Run a power flow** for the base case. Report:
   - All bus voltages (magnitude & angle)
   - All line currents (in kA, and as % of the line's assumed rating – see below)
3. **Check for undervoltage violations** (threshold: $V < 0.95$ p.u. for PQ buses).
4. **Check for overcurrent congestions** (threshold: $I > 100\%$ of line rating → overload).
5. **Determine the maximum additional active load (in MW) at bus 5** before either:
   - Voltage instability (blackout), **or**
   - Any line current exceeds **120%** of its rating, **or**
   - Any bus voltage drops below **0.90 p.u.**



Bus No	Bus Type	Bus Voltage	Generation		Load	
			MW	Mvar	MW	MVar
1	Slack	1.06	0	0	0	0
2	PV	1.01	40	---	20	10
3	PQ	---	0	0	45	15
4	PQ	---	0	0	40	5
5	PQ	---	0	0	60	10

Line	Line Impedance [Ohm]		Length [km]	Rating [kA]
	R per km	X per km		
1-2	0.02	0.06	50	1.5
1-3	0.08	0.24	60	0.5
2-3	0.06	0.25	40	0.5
2-4	0.06	0.18	30	0.5
2-5	0.04	0.12	45	0.5
3-4	0.01	0.03	55	0.5
4-5	0.08	0.24	35	0.5